Dataset Curation Strategies for Coherence Metrics in Low-Rank Models

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Introduction

Key Finding

- **No single coherence metric or validation approach is universally optimal**
- Different metrics capture different aspects of model quality
- Multi-faceted validation strategies required
- Combines mathematical rigor with practical implementation

Overview

- Mathematical foundations and statistical requirements
- Domain-specific dataset curation strategies
- Methodological guidance and implementation
- Bias detection and ethical considerations
- Reproducibility standards

Mathematical Foundations

Core Coherence Metrics

- **Pointwise Mutual Information (PMI)** and normalized variant (NPMI)
- **UMass coherence**: Co-document frequency based
- **CV coherence**: Weighted similarity measures
- Matrix factorization: $M \approx UV^T$
  - Reconstruction error minimization
  - Regularization objectives

Statistical Requirements

- **Minimum samples**: 100 per expected cluster/topic
- **Optimal datasets**: $1000+$ samples
- **Matrix condition numbers**: $< 10^{12}$
- **Sparse matrices**: $>= 1\%$ density

Computational Complexity

- SVD: $O(\min(m^2n, mn^2))$ for $m \times n$ matrices
- PMI coherence: $O(|V|^2)$ where $|V| =$ vocabulary size
- Memory limits: ~100K×80K matrices on 32GB systems

Domain-Specific Strategies

Text and NLP

Frameworks

- **Reinhart's Framework**:
  - Cohesion: Formal linking elements
  - Consistency: Logical semantic coherence
  - Relevance: Pragmatic constraints

Key Datasets

- **GCDC**: Real-world texts, 3-point coherence scales
- **CoheSentia**: GPT-generated stories, 5-point ratings
- **INSteD**: 170,000+ documents for intruder detection

Computer Vision

Visual Coherence Types

- Consistency in visual patterns
- Temporal coherence in video
- Spatial coherence in scenes

Key Datasets

- **Panda-70M**: 70.8M video clips
- **360+x**: Omnidirectional spatial coherence
- ImageNet/MS COCO adapted for coherence

Time Series Data

Temporal Coherence

- Synchronized patterns across series
- Seasonal consistency
- Causal relationships

Measurement Approaches

- Squared coherence (frequency domain)
- Cross-correlation (time domain)
- Spectral coherence (synchronization)

Scientific Datasets

Requirements

- Consistency with physical laws
- Mathematical relationships
- Domain-specific constraints

Key Resources

- **QM7-X**: 4.2M molecular structures
- **QCML**: 33.5M DFT calculations
- Materials Project

Social Networks

Structural Coherence

- Topology consistency
- Logical community grouping
- Temporal evolution patterns

Validation Approaches

- Community detection algorithms
- Centrality analysis
- Motif analysis

Methodological Guidance

Ground Truth Establishment

Approaches for Unsupervised Metrics

- **Human expert evaluation**: $\kappa >= 0.7$ agreement
- **Synthetic data**: Controlled coherence properties
- **Intrinsic metrics**: UMass, UCI coherence
- **External validation**: Word embedding similarity

Cross-Validation Strategies

Stability-Based Validation

- Temporal stability testing
- Bootstrap resampling
- Perturbation analysis
- 5-10 fold cross-validation
- Flag CV > 0.3 as unstable

Data Splitting

Temporal Data

- Forward chaining
- Chronological splits
- Prevent data leakage

General Datasets

- 60% train / 20% validation / 20% test
- Stratified splitting
- Nested cross-validation

Synthetic Data Generation

Controlled Experiments

- Matrix factorization-based generation
- Known rank structure
- Controlled noise levels
- Predetermined coherence levels

Robustness Testing

- Gradient-based perturbations
- Semantic-preserving transformations

Implementation Considerations

Technology Stack

- **DVC + Git**: Dataset versioning
- **MLflow**: Experiment tracking
- **Apache Spark**: Large-scale processing
- **Great Expectations**: Data validation
- **Gensim**: Core coherence evaluation

Scalability Guidelines

Data Size	Framework
$<$ 1GB	pandas, scikit-learn
1-100GB	Dask
$>$ 100GB	Apache Spark
ML-heavy	Ray

Quality Control

Automated Assessment

- Completeness: 95% for critical fields
- Consistency: Annotation pattern analysis
- Validity: Range checking

Human-in-the-Loop

- Flag low-confidence samples
- Consensus labeling
- Inter-annotator agreement: $\kappa > 0.8$

Bias and Ethics

Bias Detection

HBAC Algorithm

- Hierarchical Bias-Aware Clustering
- Statistical significance testing
- Unsupervised detection

Mitigation Strategies

- Data reweighting
- Subgroup balancing
- Fairness-aware sampling
- Fairness constraints in algorithms

Quantitative Metrics

- **Demographic parity**
- **Equal opportunity**
- **Disparate impact ratios**: 0.8-1.2 acceptable

Privacy Protection

- Data minimization
- Anonymization protocols
- Differential privacy
- Stakeholder consultation

Reproducibility Standards

Bronze Standard

- Code availability
- Data documentation
- Environment specification
- Random seed control

Silver Standard

- Containerization
- Automated workflows
- Comprehensive version control

Gold Standard

- Continuous integration
- Cloud deployment
- Interactive documentation
- Long-term archival

Implementation Roadmap

Phase 1: Foundation

- Basic cross-validation
- Reproducibility protocols
- Core metric implementation

Phase 2: Enhanced Testing

- Bias detection
- Synthetic data generation
- Comprehensive evaluation

Phase 3: Production Ready

- Adversarial robustness
- Cross-domain validation
- Performance optimization

Phase 4: Maintenance

- Automated monitoring
- Quality assurance systems
- Continuous improvement

Performance Thresholds

Acceptance Criteria

- **Coherence scores**:
  - UMass: > 0.4
  - UCI: > 0.3
- **Stability**: CV < 0.3
- **Cross-domain transfer**: $< 20\%$ degradation
- **Bias metrics**: Impact ratios 0.8-1.2

Future Directions

Emerging Research

- Automated bias detection systems
- Causal inference integration
- Advanced synthetic data generation
- Cross-modal coherence validation

Evolving Best Practices

- Increasing emphasis on fairness
- Interpretability requirements
- Robustness standards
- Academic-industry collaboration

Conclusion

Key Takeaways

- No universal coherence metric exists
- Domain-specific adaptation crucial
- Multi-faceted validation essential
- Balance rigor with practicality

Impact

- Enables reliable unsupervised learning
- Supports diverse domain applications
- Promotes reproducible research
- Advances coherence metric development